

KSU Capstone - Sponsor Reference Package

Integrated Production KPI Dashboard for Module Manufacturing

Purpose: Practical sponsor reference for sample data, KPI business definitions, and a conceptual dashboard direction. This is **not** a fixed design requirement; students are encouraged to propose their own analytical and visualization approach.

Reference Area	Current Status	Primary Source
Production Volume	Available	SAP production raw data
Yield	Available from grade classification	SAP production raw data (G01/G02 = Good; C = Defect)
Equipment Utilization	Available	MES equipment-status raw data
Material Consumption	Available - supporting analysis	SAP material movement / consumption raw data
Cell Loss	Under review	Dataset and detailed calculation structure to be finalized

Scope note: Read-only analytical proof of concept. Production-grade MES/SAP integration, write-back, equipment control, and automated production decisions are outside the initial scope.

1. Sample Data Guide

Dataset	Business Purpose	Typical Analysis / Dashboard Use
SAP Production Volume & Yield	Production confirmation data used to analyze actual module output and product grade.	Daily / line / model / wattage production; PCS and MW; grade mix; Yield; period and line comparison.
MES Equipment Utilization	Equipment-status data used to understand operating time and non-running conditions.	Utilization by line/equipment; RUN vs non-RUN; No WIP, Kanban Full, Tool Pause and other status trends.
SAP Material Consumption	Material movement and consumption data used to understand material usage and production flow.	Consumption by material/date/batch; production vs consumption; unusual usage patterns; supporting loss-factor investigation.
Cell Loss	Core KPI; the most appropriate raw dataset and detailed calculation structure are still being reviewed.	Placeholder only until the sponsor finalizes the source and definition.

Important: Exact field names, join keys, update frequency, and source-to-target mappings should be confirmed during requirements engineering after review of representative files.

2. KPI Business Definitions

KPI	Sponsor Business Definition	Calculation / Classification	Source
Production Volume	Actual module production for the selected period.	Aggregate production confirmation records; analyze by date, line, model, wattage and grade. PCS and MW views may both be useful.	SAP
Yield	Share of produced modules classified as good product.	G01 and G02 = Good; C = Defect. Prototype logic: $\text{Good Qty} / \text{Total Produced Qty} \times 100$. Final denominator logic to be validated.	SAP
Equipment Utilization	Share of the applicable analysis period during which equipment is in RUN status.	Prototype logic: $\text{RUN time} / \text{applicable reference time} \times 100$. Keep non-RUN statuses visible. Exact denominator to be confirmed.	MES
Cell Loss	Cell loss during module manufacturing.	Data source and detailed calculation structure are under review. Separate sample data and approved definition will follow.	TBD

Supporting metric: Material Consumption is not one of the four initial core KPIs, but it supports production-output comparison and investigation of unusual consumption.

3. Conceptual Dashboard Example - Production Overview

Illustrative values are dummy values for layout/reference only. They are not Qcells production results.



42,680 PCS
Production Volume



98.7%
Yield



84.2%
Equipment Utilization

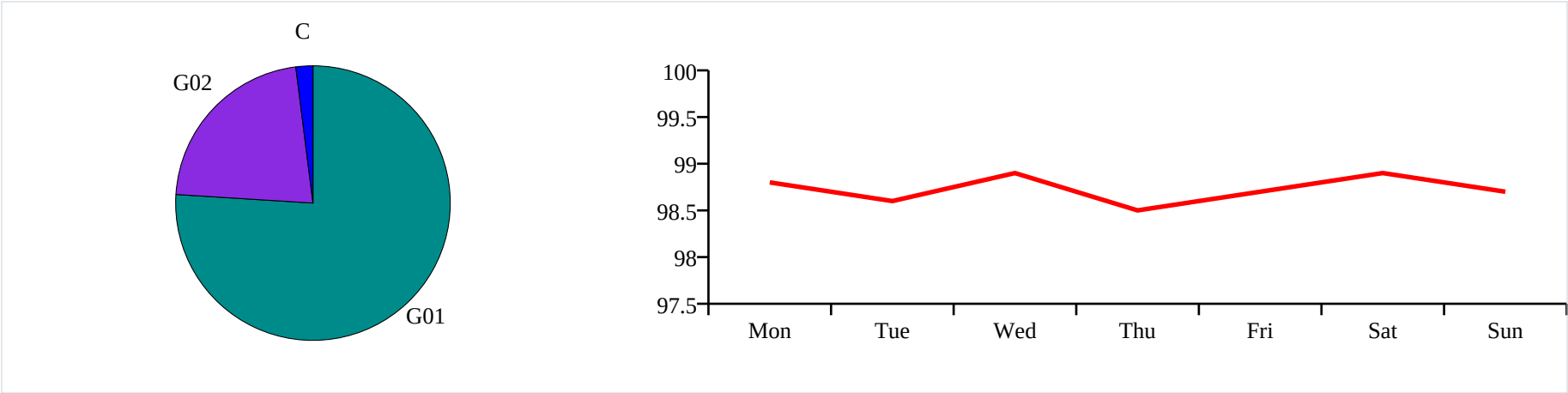


TBD
Cell Loss



4. Conceptual Dashboard Example - Production & Yield

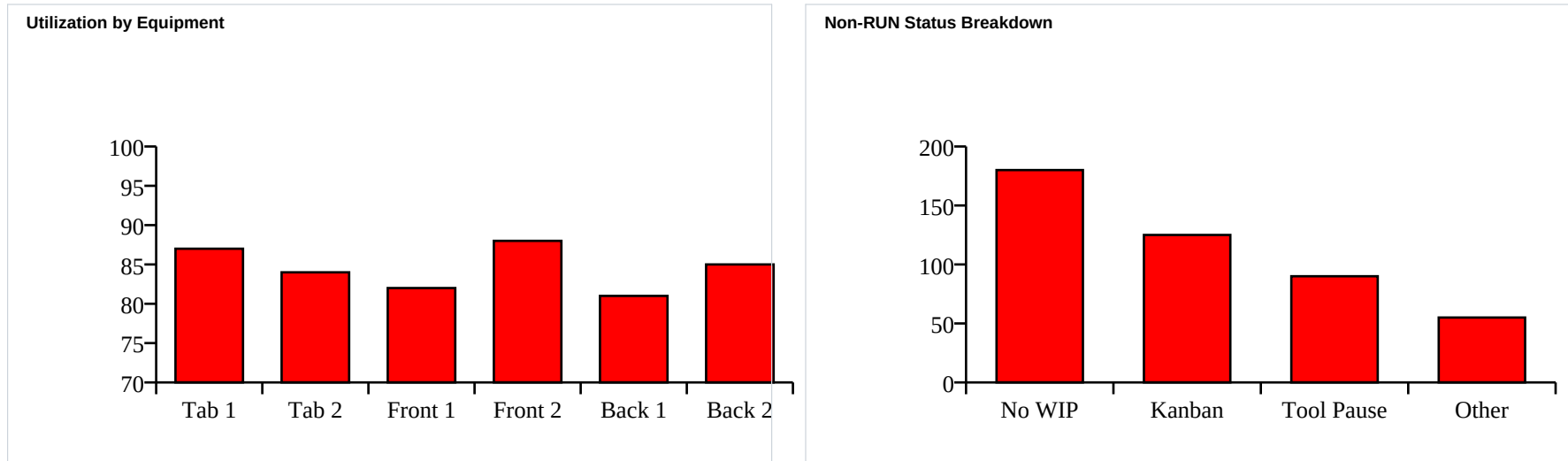
Goal: move from total output to grade mix and identify where Yield changes by line, model, or period.



View	Recommended Content
Production	PCS and MW by Date / Line / Model / Wattage
Grade Mix	G01 / G02 / C quantities and percentages
Yield Trend	Good (G01+G02) / Total Produced, trended over time
Drill-down	Line -> Model -> Wattage -> Grade -> record level, where supported

5. Conceptual Dashboard Example - Equipment Utilization

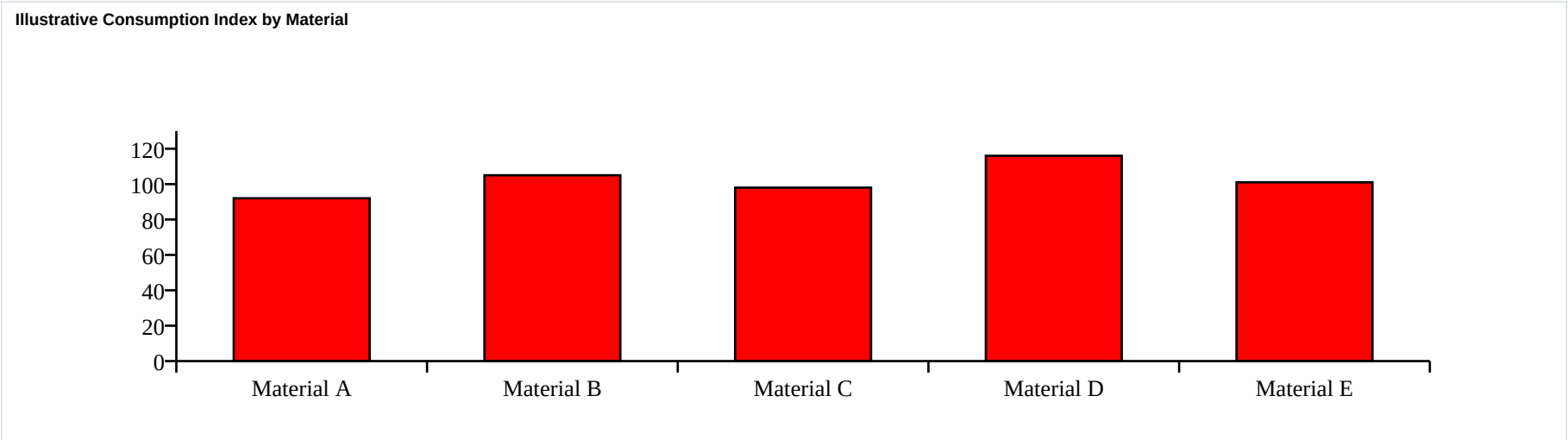
Goal: show not only utilization, but also why equipment is not in RUN status.



Recommended behavior: Selecting a low-utilization equipment item should filter the status breakdown and trend views. Keep original MES status categories visible rather than collapsing all non-RUN time into one bucket.

6. Conceptual Dashboard Example - Material Consumption

Goal: use SAP material data as supporting context for production performance and potential loss-factor investigation.



Recommended View	Purpose
Consumption Trend	Track usage by date / reporting period.
Material / Batch Detail	Review which material and batch were consumed.
Production vs Consumption	Compare production output with material usage where units and business logic allow.
Exception View	Highlight unusual changes for further investigation; do not claim causation without supporting data.

Cell Loss placeholder: Keep a reserved KPI card/page in the prototype, but label the source and calculation as under review until the sponsor selects the most useful raw dataset.

7. Student Design Freedom / Sponsor Notes

- These mock-ups are conceptual references, not mandatory layouts.
- Students may propose KPI scorecards, trend charts, line-comparison views, heat maps, exception summaries, Pareto charts, and drill-through reports.
- The dashboard should help users answer operational questions quickly: which line is underperforming, where Yield declined, which equipment has low utilization, and whether material usage aligns with production.
- Potential loss factors may be highlighted, but the dashboard should not claim root-cause causation unless the available data supports it.
- Exact fields, mappings, join keys, update frequency, and final KPI formulas should be validated during Milestone 1 with the sponsor.
- Cell Loss data source and detailed calculation structure will be provided after sponsor review.

Appendix A - Manufacturing & Data Glossary

Core production, KPI, system, and dataset terms used in the KSU capstone reference materials.

Term	Student-friendly definition / project context
KPI	Key Performance Indicator. A key metric used to evaluate production performance. KPIs can be compared with a target or threshold to identify performance gaps.
Target / Plan	The expected or planned level of production performance used as a reference for comparison with actual results.
Actual	The production result that actually occurred. Used with Target/Plan to evaluate achievement and performance gaps.
Production Volume	Actual module production output. It may be analyzed in PCS or converted to MW depending on the business view.
Yield	Percentage of produced modules classified as good product. For the provided sample: G01 and G02 = Good; C = Defect.
Grade	Product quality classification used in the SAP production data, such as G01, G02, and C.
Model	The product model being manufactured. Different module products are managed and analyzed by model.
Wattage / Power Class	The electrical output class of a module, used to group and analyze production by power level.
PCS	Pieces. The number of modules produced.
MW	Megawatts. A production-volume view expressed as electrical capacity rather than only module count.
MES	Manufacturing Execution System. A shop-floor system used to manage or record production, process, and equipment information.
SAP	Enterprise resource planning system used for production confirmation, material movement/consumption, inventory, and related business records.
Production Order	A production instruction/order defining what product is to be produced and the related production activity.
Line	A manufacturing line, such as R1, R2, or R3, used as a primary dimension for production-performance analysis.
Process	A manufacturing step or stage within the overall module-production flow.
WIP	Work in Process. Material or product that has entered production but is not yet a completed finished module.

Appendix A - Manufacturing & Data Glossary (continued)

Term	Student-friendly definition / project context
Raw Data	Source-level data before dashboard aggregation or analytical transformation.
Unified Dataset	An analytical dataset created by combining relevant data from sources such as MES, SAP, and Excel.

Note: These definitions are intended to help students understand the sample data and project context. Source-system-specific meanings should be confirmed with the sponsor when needed.

Appendix A - Manufacturing & Data Glossary

Equipment, material, analysis, and data-engineering terms used in the KSU capstone reference materials.

Term	Student-friendly definition / project context
Equipment Utilization	Percentage of the applicable analysis time during which equipment is in RUN status. The final denominator should be validated with the sponsor.
RUN	MES equipment status indicating that the equipment is operating.
Non-RUN	General analytical grouping for time when equipment is not in RUN status. Original MES status categories should remain available for detail analysis.
No WIP	A non-RUN condition indicating that work-in-process/input material is not available for the equipment to continue production.
Kanban Full	A condition in which downstream/buffer capacity is full, preventing normal flow to the next stage.
Tool Pause	An MES equipment status representing a temporary pause condition. Detailed interpretation should follow the source-system definition.
Downtime	Time during which equipment is not operating. Downtime should be analyzed by the available status/reason categories.
Tabber	Equipment/process that connects solar cells into strings during module manufacturing.
Material Consumption	Material quantity used or posted during production. SAP material data can support production-versus-consumption and trend analysis.
Material Number	SAP identifier/code for a material.
Batch / Lot	Identifier used to trace a specific material or production lot/batch.
Storage Location	SAP location code representing where material is stored or managed in inventory.
Movement Type	SAP code that identifies the business meaning/type of a material movement or posting.
Cell Loss	Loss of cells during module manufacturing. The project data source and detailed calculation structure are currently under review.
Loss Factor	A factor potentially associated with production loss or performance deterioration. It should not be treated as proven root cause without supporting evidence.
Trend Analysis	Analysis of how a KPI or metric changes over time.

Appendix A - Manufacturing & Data Glossary (continued)

Term	Student-friendly definition / project context
Pareto Chart	A chart that ranks categories from largest to smallest to help identify the most significant contributors.
Drill-down	Moving from a high-level KPI into more detailed dimensions, for example Line -> Model -> Wattage -> Grade.
Data Mapping	Definition of how fields or business concepts from different source systems correspond to one another.
Join Key	A field or combination of fields used to connect records from different datasets.
Data Validation	Checking data for issues such as missing values, duplicates, inconsistent formats, or invalid records.
Dashboard Refresh	Updating dashboard data and calculations when new source data becomes available.
POC	Proof of Concept. A prototype used to demonstrate that the analytical approach is feasible and useful before production implementation.